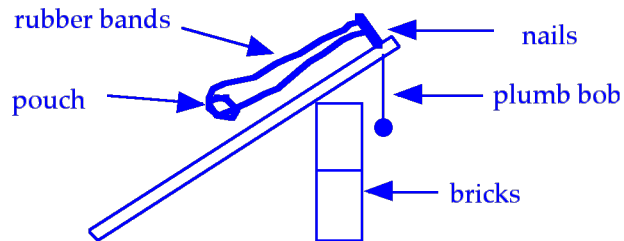


YEAR 12 PHYSICS
PROJECTILE MOTION INVESTIGATION

AIM. To determine the mathematical relationship between the angle of release and the range of the projectile.

METHOD

1. Use a launcher as shown below to release a marble.



Measure the angle of inclination using the plumb bob and a protractor.

2. List the variables that must be kept constant during the experiment.
3. Vary the angle of inclination from 20° through to 80° in 10° increments. Record the angle and distance covered horizontally by the marble in a suitable table.
4. For **one trial** (at an angle between 30° and 60°), roughly calculate the maximum height achieved in the flight by using an inclinometer at right angles to the flight path.

Use this information to calculate the approximate speed of the projectile at release.

5. Use your calculator to investigate the possible relationship between the angle of projection and the range achieved. Give consideration to trigonometric functions such as sine, cosine and tangent for example.

RESULTS

Draw up suitable tables. Any calculations made should be in this section.

CONCLUSION

This must answer the aim based on your results and observations.

CRITICAL ANALYSIS

List all possible sources of error and/or approximations that had to be made during the experiment.

Make suggestions on how to improve the experiment.

QUESTIONS

1. Which angle gave the greatest range? Was this the expected result? Explain your answer.
2. Working from first principles, derive an equation that relates the range, speed of release and angle of release for any object projected at an angle.